Rishabh Singh

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EDUCATION

Vellore Institute of Technology (VIT) — **Oct 2022 – May 2027**
Integrated M.Tech in Artificial Intelligence — **CGPA: 8.59 / 10.0**

TECHNICAL SKILLS

Programming Languages: C++, Python, SQL, R
Machine Learning and Deep Learning: TensorFlow, PyTorch, Keras, Scikit-learn, Detectron2, OpenCV
Data Analysis and Visualization: Pandas, NumPy, Matplotlib, Tableau, Power BI
Databases: MySQL, MongoDB
MLOps and Development Tools: Git, GitHub, Docker, Postman, FastAPI, TypeScript, AWS, LangChain, N8N
Core CS Concepts: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, SDLC

PROJECTS

Panoptic Image Segmentation
Technologies: Python, PyTorch, Detectron2, TensorFlow, Keras, OpenCV, CUDA, NumPy

- Architected an end-to-end panoptic segmentation pipeline using Detectron2, training on 12,000+ labeled images to reduce GPU memory overhead by 22% via optimized tensor operations.
- Accelerated inference speed by 28% by implementing parallel data loading and efficient batching strategies, enabling real-time processing of video streams at 30 FPS.
- Deployed production-ready inference system achieving 82% segmentation accuracy across diverse test scenarios, processing 1,200+ samples with sub-second latency.

Self-Learning Recommendation System
Technologies: Python, NumPy, Pandas, Scikit-learn, SciPy, Flask, REST APIs, MySQL

- Engineered a collaborative filtering engine utilizing matrix factorization, reducing recommendation computation time by 38% through sparse matrix operations and vectorization.
- Optimized a data pipeline to process 6,500+ user-item interactions, decreasing memory usage by 42% and improving preprocessing throughput by 3x.
- Formulated a scalable Flask REST API serving personalized recommendations, maintaining sub-200ms response times under 60+ concurrent users with 91% recommendation relevance.

AURA – AI-Powered Service Center Dashboard
Technologies: Python, FastAPIs, TypeScript, React, Recharts, Data Visualization

- Designed a real-time analytics dashboard tracking 12 operational KPIs with automated anomaly detection, decreasing manual monitoring effort by 45% across service operations.
- Integrated 6 ML inference pipelines into an interactive dashboard with role-based access for 4 user types, displaying confidence scores and trend analysis through 18 reusable components.
- Streamlined a data processing pipeline handling 650+ events daily with sub-80ms latency, enabling continuous monitoring and cutting decision response time by 40%.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate – Amazon Web Services (June 21, 2026 - June 21, 2029) Link
- Programming Fundamentals – Duke University, Coursera (Sep 2023) Link
- Applied Machine Learning – Coursera (Dec 2023) Link
- Cloud Computing – NPTEL, IIT (May 2024) Link
- Data Science for Beginners – NASSCOM (Mar 2025) Link

LEADERSHIP

- **Google Summer of CodeFest 2025 – Innovators Club, VIT:** Secured a position in the Development Round among 100+ participants by formulating a comprehensive machine learning project proposal, ranking in the top 20% of applicants.
- **MERN X AI Hackathon – Mern Matrix Club:** Spearheaded the architecture of an AI-powered full-stack web application integrating 3+ third-party machine learning APIs within a 24-hour build window to qualify for the Development Round.
- **Event Management Lead – Cooking and Feasting Club, VIT:** Orchestrated online and offline community events by directing a cross-functional team of 15+ volunteers to execute 15+ events for 200+ participants, elevating the task delivery rate by 20% through structured sprint planning.